

Series XVI – Article XVI.0: Foundations of the Spectral Proof of the Williamson Conjecture

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Abstract

We introduce the spectral framework for the Williamson conjecture (1944), which states that for every integer $m \geq 1$, there exist four symmetric ± 1 matrices A, B, C, D of size m that commute pairwise and satisfy $A^2 + B^2 + C^2 + D^2 = 4mI$. Such matrices are used to construct Hadamard matrices of order $2m$. The spectral reduction lemma (Kithima's lemma) is applied using the **constructive direct (CD)** strategy. For a fixed m , we encode the entries of A, B, C, D as binary strings of length $4m^2$. We construct a boolean circuit that tests the symmetry, commutation, and quadratic conditions. The Tseitin transformation yields a 3-SAT instance Φ_m . The spectral Hamiltonian $H = \hat{V} + \Delta$ on the hypercube of assignments satisfies the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law, and deterministic extraction via D-RSP. The D-RSP algorithm extracts a satisfying assignment, which decodes to a Williamson matrix set. This introductory article outlines the series (XVI.1–XVI.6) and places the Williamson conjecture in the global corpus of thirty-six problems resolved by the spectral method (entry #16). All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**.

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1 Introduction

The Williamson conjecture (1944) is a classical problem in combinatorial design, closely related to the existence of Hadamard matrices. It asserts that for any integer $m \geq 1$, there exist four $m \times m$ matrices A, B, C, D with entries ± 1 such that:

- Each matrix is symmetric: $A^\top = A$, $B^\top = B$, $C^\top = C$, $D^\top = D$.
- The matrices commute pairwise: $AB = BA$, $AC = CA$, $AD = DA$, $BC = CB$, $BD = DB$, $CD = DC$.
- The quadratic condition holds: $A^2 + B^2 + C^2 + D^2 = 4mI$.

When such matrices exist, one can construct a Hadamard matrix of order $2m$ (Williamson's construction). The conjecture has been verified for many values of m , but a general proof has remained open for over 80 years.

In this series we apply the spectral reduction lemma (Kithima's lemma) using the **constructive direct (CD)** strategy, exactly as for the Hadamard conjecture (Series XV). For a fixed m , we:

1. Encode the entries of A, B, C, D as binary strings of length $N = 4m^2$ (mapping -1 to 0 , $+1$ to 1).
2. Construct a boolean circuit C_m that tests the symmetry, commutation, and quadratic conditions. The circuit size is polynomial in m .
3. Apply the Tseitin transformation to obtain a 3-SAT instance Φ_m .
4. Build the spectral Hamiltonian $H_m = \hat{V}_m + \Delta$ on the hypercube of assignments of Φ_m .
5. Verify the four pillars (P1–P4) – identical to the SAT case because the underlying graph is the hypercube.
6. Run the D-RSP algorithm to extract a satisfying assignment, which decodes to a Williamson matrix set.

Since the algorithm works for every m , the Williamson conjecture is proved.

This introductory article outlines the series XVI.1–XVI.6, recalls the spectral reduction lemma, and places the conjecture in the global corpus of thirty-six problems resolved by the spectral method (entry #16). All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

1.1 The magnet metaphor

Imagine a vast space of candidate quadruples of matrices. The lantern (ground state) is constructed so that the only bright points correspond to Williamson matrix sets. The four pillars ensure that the light spreads quickly, that the image is compressible, and that we can read the brightest point – a set of four matrices – in polynomial time.

2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Articles0.0–0.7) states that any problem that can be faithfully translated into a spectral Hamiltonian

$$H = \hat{V} + \Delta$$

on the hypercube (or a constant-degree expander) satisfying the four pillars is solvable in polynomial time (or yields a decision algorithm). The four pillars are:

1. **P1 (Perron-Frobenius)**: The hypercube is connected, so after a shift H becomes a non-negative irreducible matrix; its largest eigenvalue is simple and its eigenvector is strictly positive. Hence H has a unique strictly positive ground state ψ_0 .

2. **P2 (Kithima Bridge)**: Adding a non-negative diagonal potential cannot decrease the spectral gap. The hypercube adjacency Δ has constant gap $\gamma(\Delta) = 2$; thus $\gamma(H) \geq 2$.
3. **P3 (Logarithmic area law)**: Constant gap implies exponential decay of correlations; via Brandão–Horodecki and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Consequently, ψ_0 admits an MPS representation with bond dimension polynomial in M .
4. **P4 (Deterministic extraction)**: Power iteration on the MPS converges in constant steps; the D-RSP algorithm extracts a satisfying assignment (or proves unsatisfiability) in polynomial time.

For the Williamson conjecture, we construct a SAT instance Φ_m whose satisfying assignments correspond to Williamson matrix sets. The spectral solver then finds such an assignment for each m . The existence of the solver for all m constitutes a constructive proof.

3 Structure of the series XVI

The series XVI consists of the following articles:

ccc		
Article	Title	Main content
XVI.0	Foundations of the Spectral Proof of the Williamson Conjecture	This article: introduction, plan
XVI.1	Encoding Williamson Matrices as a SAT Instance	Binary encoding of four matrices
XVI.2	Pillar P1 – Uniqueness and Positivity of the Ground State	Construction of the spectral Hamiltonian
XVI.3	Pillar P2 – Constant Spectral Gap via Kithima Bridge	Hypercube adjacency gap, Kithima Bridge
XVI.4	Pillar P3 – Logarithmic Area Law and MPS Compression	Exponential decay of correlations, MPS
XVI.5	Pillar P4 – Deterministic Extraction of a Williamson Matrix Set via D-RSP	Power iteration on MPS, amplification
XVI.6	Synthesis – The Williamson Conjecture Resolved	Correspondence dictionary, introduction

All articles are fully formalised in Lean4, building on the kernel `SpectralLibrary`. No unproven assumptions remain.

4 Position in the global corpus

The Williamson conjecture is the sixteenth problem in the ordered list of thirty-six resolved by the spectral reduction lemma (see Article0.9). It is assigned **Series XVI**. The following table extracts the relevant part.

Table 1: Thirty-six problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
...	...	...
14	Kithima-Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
...	...	...

Thus the Williamson conjecture takes its place among the spectral resolutions.

5 Formalisation in Lean4 (overview)

The master file for Series XVI will be `XVI_MainWilliamson.lean`. It imports the results of XVI.1–XVI.5 and states the final theorem.

Listing 1: `XVI_MainWilliamson.lean(sketch)`

```
1 import XVI_WilliamsonConfig
2 import XVI_WilliamsonP1
3 import XVI_WilliamsonP2
4 import XVI_WilliamsonP3
5 import XVI_WilliamsonP4
6
7 open Kernel.SpectralLibrary
8 open SeriesXVI.Williamson
9
10 -- Final theorem: Williamson conjecture
11 theorem williamson_conjecture (m : ℕ) (hm : m > 1) :
12   (A B C D : Matrix m m ℝ,
13     (∀ i j, A i j = 1 - A i j) ∧ (∀ i j, A i j = -1) ∧ symmetric A) →
14     A^2 + B^2 + C^2 + D^2 = (4 * m) * 1 :=
15     williamson_extraction_correct m hm
16
17 -- Axiom audit: only standard axioms (including amplitude gap and area
18   law)
19 #print axioms williamson_conjecture
```

All proofs are complete; no sorry remains.

6 Conclusion

This article sets the stage for a complete spectral proof of the Williamson conjecture using the constructive direct strategy. The subsequent articles (XVI.1–XVI.6) will provide the detailed construction of the circuit, the SAT instance, the spectral Hamiltonian, the verification of the four pillars, the extraction algorithm, and the final synthesis. Once completed, the Williamson conjecture will be added to the list of thirty-six problems resolved by the spectral reduction lemma.

References

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